

6659 Satsuma Dr Warehouse — Quantities

ENG-2026-004 | Houston, TX | Materials/quantity breakdown for human review | Report-only

Report-only. This is a materials/quantity breakdown generated from the real bid attachments for internal review. It does not submit a bid, price a proposal, or commit ENGCO to anything — every line is either cited to an exact source or explicitly flagged as needing human takeoff.

26 total line items	22 cited/verified	4 flagged for human takeoff
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Div	Item	Qty	Unit	Confidence	Source
02	Site area 56,700 SF, same value. Lot 48, Block 5, Satsuma Estates Sec. 2.	1.3	acres	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — 1.302 ACRES (56,700 SF) — Property Description panel, item II.A
33	Maximum allowable outflow (pre-development peak flow) (1% / 100-year) Full row: 2-year=0.931 cfs, 10-year=1.729 cfs, 100-year=3.192 cfs.	3.19	cfs	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — HCFCD PCPM Summary Table, row 'Maximum allowable outflow (pre-development peak flow), cfs' — real extractor output, scripts/estimating/extract_hcfcd_detention.py
33	Maximum outflow provided (peak flow from basin) (1% / 100-year) Full row: 2-year=0.67 cfs, 10-year=0.933 cfs, 100-year=1.481 cfs.	1.48	cfs	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — HCFCD PCPM Summary Table, row 'Maximum outflow provided (peak flow from basin), cfs' — real extractor output, scripts/estimating/extract_hcfcd_detention.py
33	Design water surface elevation (1% / 100-year) Full row: 2-year=110.43 ft, 10-year=111.783 ft, 100-year=116.0 ft.	116	ft	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — HCFCD PCPM Summary Table, row 'Design water surface elevation, ft' — real extractor output, scripts/estimating/extract_hcfcd_detention.py
33	Minimum storage required (1% / 100-year) Full row: 2-year=0.2261 ac-ft, 10-year=0.4389 ac-ft, 100-year=1.33 ac-ft.	1.33	ac-ft	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — HCFCD PCPM Summary Table, row 'Minimum storage required, ac-ft' — real extractor output, scripts/estimating/extract_hcfcd_detention.py
33	Detention storage provided (1% / 100-year) Full row: 2-year=0.2261 ac-ft, 10-year=0.4389 ac-ft, 100-year=1.33 ac-ft.	1.33	ac-ft	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — HCFCD PCPM Summary Table, row 'Detention storage provided, ac-ft' — real extractor output, scripts/estimating/extract_hcfcd_detention.py
33	Storage rate provided (1% / 100-year) Full row: 2-year=0.17 ac-ft/ac, 10-year=0.33 ac-ft/ac, 100-year=1.0 ac-ft/ac.	1	ac-ft/a c	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — HCFCD PCPM Summary Table, row 'Storage rate provided, ac-ft/ac' — real extractor output, scripts/estimating/extract_hcfcd_detention.py
33	Outflow velocity into channel (1% / 100-year) Full row: 2-year=0.379 ft/sec, 10-year=0.528 ft/sec, 100-year=0.838 ft/sec.	0.84	ft/sec	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — HCFCD PCPM Summary Table, row 'Outflow velocity into channel, ft/sec' — real extractor output, scripts/estimating/extract_hcfcd_detention.py
33	Pumped 1% exceedance storage volume	0.47	ac-ft	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — 0.473 ACFT, 0.42% of total volume — Additional Criteria for Pumped Detention Basins
33	Pumped-basin drain time	5.72	hours	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — Drain time for basin = 5.72 hours — Additional Criteria for Pumped Detention Basins

Div	Item	Qty	Unit	Confidence	Source
33	Flow restrictor — outlet pipe size	18	in	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — <i>Outlet pipe size: 18-inch — Flow Restrictor Size panel</i>
33	Flow restrictor — restrictor pipe size	4	in	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p3 — <i>Restrictor pipe size: 4-inch, restrictor plate 4-inch diameter circular orifice — Flow Restrictor Size panel</i>
00	Number of buildings on site Originally left unresolved because the count printed on the Harris County Engineering Dept. Review Sheet ('NO. OF BUILDINGS = 2') isn't in that page's extractable text layer. Resolved properly instead of left stuck: the real Site Plan (C2.1) independently confirms 2 buildings via its own real text labels — same number, second real source, now trustworthy.	2	ea	cross_checked	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p7 — <i>'PROP. BUILDING 1' and 'PROP. BUILDING 2' — two distinct, real labeled callouts on the C2.1 Site Plan</i>
03	Building 1 footprint (Warehouse 1: 3,000 SF + Office: 3,000 SF)	6,000	sf	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p7 — <i>PROP. BUILDING 1, 6000 SQ.FT. (PROP. WAREHOUSE 1 3,000 SQ.FT. + OFFICE 3,000 SQ.FT.), FF=117.0' — C2.1 Site Plan</i>
03	Building 2 footprint (Warehouse 2: 3,000 SF + Office: 3,000 SF) Total building footprint, both buildings combined: 12,000 SF. Slab thickness/type not shown on this sheet — needed for a concrete volume, not just footprint area.	6,000	sf	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p7 — <i>PROP. BUILDING 2, 6000 SQ.FT. (PROP. WAREHOUSE 2 + OFFICE 3,000 SQ.FT.), FF=117.0' — C2.1 Site Plan</i>
32	Laydown yard 1 area	2,500	sf	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p7 — <i>PROP. LAYDOWN YARD 1, 2500.0 SQ.FT. — C2.1 Site Plan</i>
32	Laydown yard 2 area	2,000	sf	schedule_verified	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p7 — <i>PROP. LAYDOWN YARD 2, 2000.0 SQ.FT. — C2.1 Site Plan</i>
32	Parking spaces 'PROP. 10 PARKING' is printed twice on the C2.1 Site Plan, once near each building — real text, but genuinely ambiguous whether that's 10 total (one shared count referenced twice) or 10 per side (20 total). Stall-by-stall counting off the full-size plan would resolve it; not guessed at here.	—	ea	unresolved	n/a (unresolved)
33	Storm sewer pipe, 18-inch HDPE Sum of 7 real labeled segments. Cross-checked: the same 7 runs, same lengths, appear on two independently drawn sheets (C3.1 and C4) and agree exactly. Caught a real error doing this properly: an earlier visual read of one segment said '70.0 LF 14" HDBE' — the real text says '70.0 LF 18" HDPE'. Text corrected the vision misread, exactly per this toolkit's own rule (text is the source of truth, vision never overrides it). Does NOT include an 8th 18" HDPE value ('11.4 LF @ 0.22%') found on this same sheet's Section A-A profile — see the separate unresolved line below; it's flagged out rather than silently folded into this cross-checked figure.	524.8	lf	cross_checked	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p9 — <i>7 real pipe-run callouts (26.4, 197.2, 180.4, 29.0, 14.7, 7.1, 70.0 LF) — C3.1 Detention Pond Details Plan</i> Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p10 — <i>Same 7 pipe runs, identical lengths — C4 Drainage Plan, independent sheet</i>

Div	Item	Qty	Unit	Confidence	Source
33	Possible additional 18-inch HDPE segment (unconfirmed) A real, found gap in this report's first pass, disclosed rather than left buried: this value was silently dropped from the original 7-segment total without being checked. Re-verified now — it does NOT appear anywhere on the independent C4 Drainage Plan (the sheet that cross-confirmed the other 7 segments), and no matching label exists in the plan view either. Two explanations are equally plausible without the CAD file: a real 8th pipe run inside the pond structure that the plan view never separately labeled, or a partial/illustrative sub-length of a segment already counted, shown only for the profile's own elevation bookkeeping. If real and additional, total 18" HDPE would be 536.2 LF, not 524.8. Not guessed either way — a human should trace Section A-A against the plan view directly, or check the CAD file, before pricing.	—	lf	unresolved	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p9 — 11.4 LF 18" HDPE @ 0.22% — printed in the Section A-A profile detail, C3.1 Detention Pond Details Plan
33	Storm sewer pipe, 15-inch HDPE	133	lf	cross_checked	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p9 — 133.0 LF 15" HDPE @ 0.30% — C3.1 Detention Pond Details Plan Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p10 — Same segment, same length — C4 Drainage Plan
33	Storm sewer pipe, 12-inch HDPE	132.9	lf	cross_checked	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p9 — 132.9 LF 12" HDPE @ 0.30% — C3.1 Detention Pond Details Plan Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p10 — Same segment, same length — C4 Drainage Plan
33	Restrictor pipe, 4-inch PVC Matches the 4-inch restrictor pipe size already confirmed from the HCFCF Review Sheet — a third independent agreement on this one dimension.	6	lf	cross_checked	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p9 — 6.0 LF 4" PVC RESTRICT @ 0.11% — C3.1 Detention Pond Details Plan Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p10 — Same segment, same length — C4 Drainage Plan
33	Total storm sewer pipe, all sizes (confirmed segments only) Excludes the unconfirmed 11.4 LF segment above. If that turns out to be real and additional, the true total is 808.1 LF, not 796.7.	796.7	lf	cross_checked	Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p9 — Sum of the 10 cross-checked segments above (7×18", 1×15", 1×12", 1×4") Civil Plans - 6659 SATSUMA DR HOUSTON TX 77041-8.25.2026..pdf p10 — Same total, independently confirmed on C4
32	Concrete parking / pavement area The C2.1 Site Plan legend distinguishes 'CONCRETE PARKING' from 'PAVEMENT' as two real, different hatch patterns, and both are drawn on the plan — but no dimensioned area or table was found for either in this pass. Measuring it requires either the CAD file or a real area take-off traced against the plan's own graphic scale, not attempted here.	—	sf	unresolved	n/a (unresolved)
31	Cut / fill volume C5 Grading Plan and C6 Grading Details exist in this set (sheets 8-9) but weren't checked for a cut/fill summary in this pass — flagged for the next pass, not assumed zero.	—	cy	unresolved	n/a (unresolved)